

Homework 1:

Il Problema della Variabile Omessa

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UNIVERSITÀ
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Da un secolo, oltre.

Panoramica

Obiettivo:

Osservare l'influenza di una
covariata importante (β_2)
all'interno di un modello
statistico

Procedimento:

- DGP
- Scenari e Monte Carlo
- Convergenza e Bias

DGP

$$y_i = \beta_0 + \beta_1 x_{(1,i)} + \beta_2 x_{(2,i)} + \varepsilon_i$$

$$\varepsilon \sim N(0,1)$$

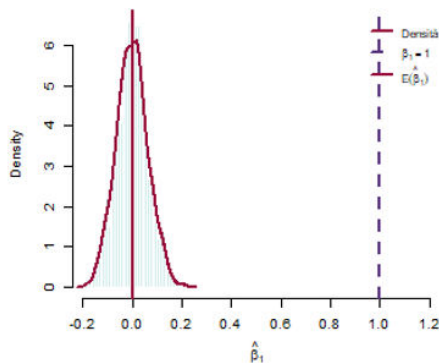
$$x_1, x_2 \sim N_2(\underline{0}, \Sigma)$$

Risultati Ottenuti

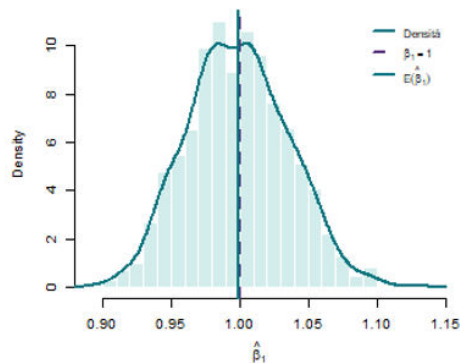
Scenario 1:

x_1 e x_2 sono correlate, $\rho = 0.5$

Scenario 1: Omitted Variable



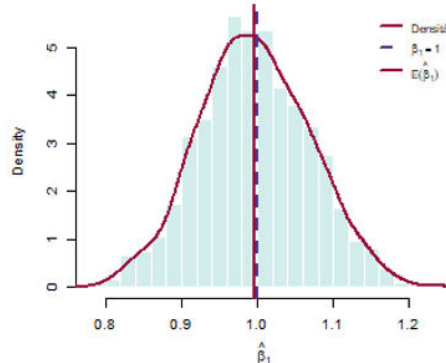
Scenario 1: Full Model



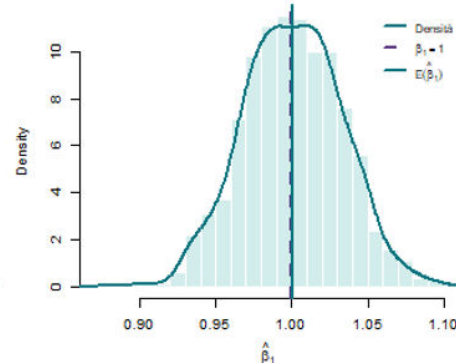
Scenario 2:

x_1 e x_2 sono indipendenti, $\rho = 0$

Scenario 2: Omitted Variable



Scenario 2: Full Model

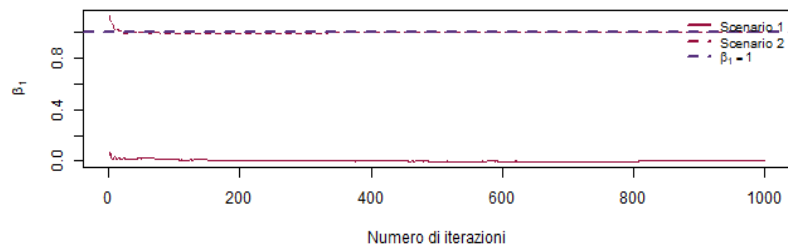


$$Bias(\widehat{\beta}_1) = \beta_2 \frac{Cov(x_1, x_2)}{Var(x_1)}$$

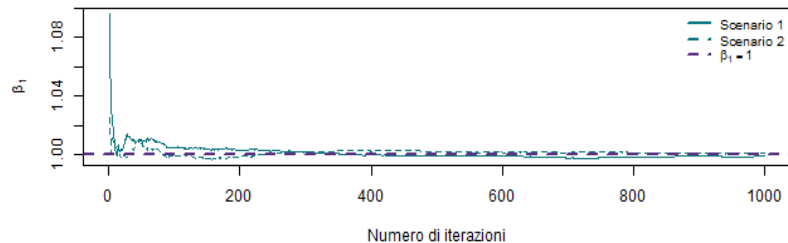
Ulteriori casi di studio

Convergenza con Monte Carlo

Convergenza: Modello OMESSO

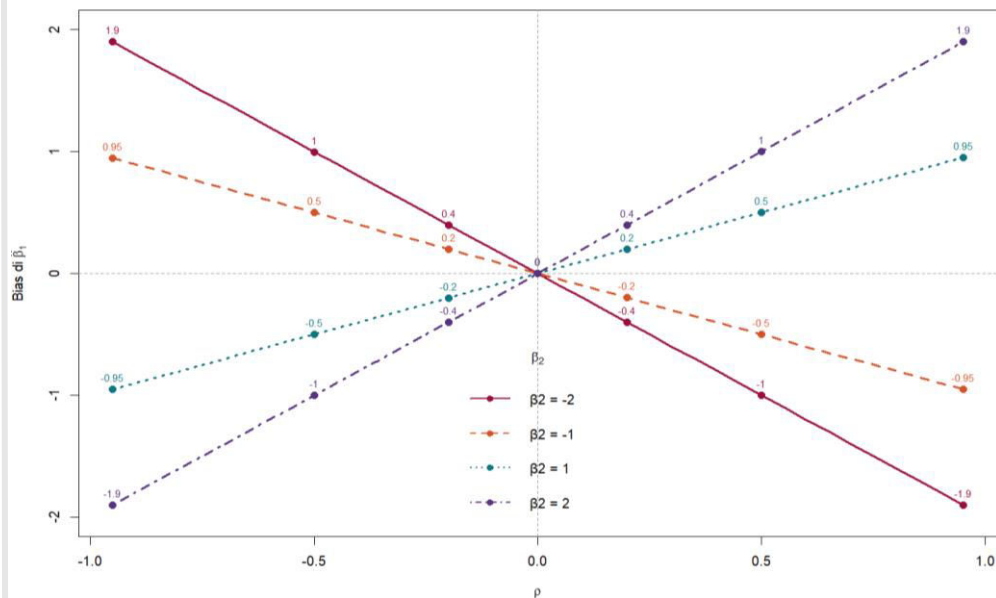


Convergenza: Modello Completo



Bias al variare di ρ e β_2

Bias di $\hat{\beta}_1$ al variare di ρ per diversi β_2



References

- Slides di Statistical Learning fornite dalla docente
- Lab R: Monte Carlo Simulations fornito dalla docente